Felipe Espreafico Guelerman Ramos

— Curriculum Vitae —

4 place Jussieu – 75005 Paris – France
espreafico-guelerman@imj-prg.fr • espreafico-guelerman.github.io/

Presentation and Research Interests

- I am currently a postdoctoral fellow at the Institut de Mathématiques de Jussieu of the Sorbonne University in Paris, under the supervision of Professor Penka Georgieva, funded by her ERC grant ROGW 864919.
- Research Interests: Enumerative Geometry, Singularity Theory, Motivic Homotopy theory, Mathematical Physics (hep-th)
- I work in Algebraic Geometry in a very broad sense. I'm interested in understanding the core structure of numerical invariants, especially over fields which are not the complex numbers and how that relates to enumerative geometry, birational geometry, singularity theory and mathematical physics. I am also fascinated by Mirror Symmetry and what we can learn about enumerative invariants from Hodge theory.
- Concretely, I've been working on modularity of real Gromov-Witten theory, refined Donaldson-Thomas invariants and motivic homotopy theory.

Employment

Postdoctoral Fellow — **TU Darmstadt, Darmstadt**
DFG Walter Benjamin Position — *2026–2028*
Supervision: Sabrina Pauli

Postdoctoral Fellow — **Sorbonne University, Paris**
Funded by the European Research Council (ROGW 864919) — *2024–2026*
Supervision: Penka Georgieva

Education

PhD in Mathematics — **Rio de Janeiro**
Institute of Pure and Applied Mathematics, with a year (2022-2023) spent at University of Heidelberg — *2020–2024*
Supervisors: Hossein Movasati and Johannes Walcher

BSc in Mathematics — **São Carlos**
University of São Paulo, Final Grade 9.6/10 — *2016–2019*
with a semester (2018-2019) spent at Leibniz University Hannover

Research Experience and Grants Awarded

Postdoctoral Level

Curve Counting via Quadratic Localizaton and Mirror Symmetry — **TU Darmstadt**
DFG Walter Benjamin Position, Darmstadt — *2026–2028*

Graduate Level

Refinements of Enumerative Invariants in Physics — **University of Heidelberg**
CAPES PhD Internship Abroad Grant, Heidelberg — *2022–2023*
Supervisor: Johannes Walcher

Open Gromov-Witten invariants and moduli of enhanced Clalabi-Yau threefolds — **IMPA**
CNPq PhD Fellowship (4-year), Rio de Janeiro — *2020–2024*
Supervisor: Hossein Movasati

Undergraduate Level

Intersection Homology and Applications to Singularity Theory — **University of São Paulo**
FAPESP Undergraduate Research Grant, São Carlos — *2019–2019*
Supervisor: Prof Nivaldo Grulha.

Tjurina Transgform and Determinantal Singularities. — **Leibniz Universität Hannover**
FAPESP Research Internship Abroad Grant, Hannover — *2018–2019*
Supervisor: Prof Anne Frühbis-Krüger.

Introduction to Analytic Geometry — **University of São Paulo**
FAPESP Undergraduate Research Grant, São Carlos — *2017–2018*

Supervisor: Prof Nivaldo Grulha and Prof Dr Maria Aparecida Soares Ruas.

Transcendental Methods of algebraic/complex geometry in hyperbolic geometry — **University of São Paulo**
FAPESP Undergraduate Research Grant, São Carlos — *2016–2017*
Supervisor: Prof Alexandre Ananin.

Prizes and Awards

Outstanding Academic Performance as Undergraduate student
University of São Paulo — *2019*

Honorable mention
Brazilian Mathematics Olympiad (*undergraduate level*) — *2017*

Bronze Medal
Brazilian Chemistry Olympiad (*high school level*) — *2015*

Silver Medal
São Paulo's Chemistry Olympiad (*high school level*) — *2015*

Bronze Medal
Brazilian Physics Olympiad (*high school level*) — *2015*

Teaching and Organizational Experience

Coorganizer of the IMPA Student Seminar — **Rio de Janeiro**
Inst. of Pure and Appl. Math. — *Aug 2023-Feb 2024*

Teaching Assistant for Riemann Surfaces — **Rio de Janeiro**
Inst. of Pure and Appl. Math. — *Aug-Dez, 2023*

Teaching Assistant for Differential Topology — **Rio de Janeiro**
Inst. of Pure and Appl. Math. — *Jan - Mar, 2022*

High School teaching — **São Carlos**
Voluntary teacher at a preparatory course for High School Students — *2017–2019*

Selected Conferences

New Developments in Singularity Theory — **Miami**
Institute for Mathematical Sciences of the Americas, Invited Speaker — *2025*

Real Algebraic Geometry Days — **Rennes**
Inst. of Math. Research of Rennes, Invited Speaker — *2025*

Atelier Real and Motivic Geometry — **Le Croisic**
Domaine de Port aux Rocs, Speaker — *2025*

Theory of Atoms, Educational Workshop — **Sofia**
International Centre for Mathematical Sciences, Invited Speaker — *2024*

Motivic homotopy in interaction — **Marseille**
Centre International of Mathematical Meetings, Speaker — *2024*

Hodge theory, Mirror Symmetry, and Physics of Calabi-Yau Moduli — **Heidelberg**
University of Heidelberg, Invited Speaker — *2023*

Structures in Enumerative Geometry — **Sheffield**
University of Sheffield, Participant — *2023*

International School on Singularities and Lipschitz Geometry — **Cuernavaca**
Autonomous National University of Mexico, Participant — *2018*

Invited Lectures

Counting with Quadratic Forms — **Brigham Young University**
Lecture at the Algebraic Geometry Seminar, Provo — *2025*

Refined Donaldson-Thomas Invariants and Milnor Fibres — **Inst. of Math. Sciences of the Americas**
Lecture at the Conference New Developments on Singularity Theory, Miami — *2025*

On Motivic and Arithmetic Donaldson-Thomas invariants — **Inst. of Math. Research of Rennes**
Lecture at the Conference Real Algebraic Geometry Days, Rennes — *2025*

Counting Lines on Hypersurfaces over General Fields — **University of Bologna**
Lecture at the Bologna Algebraic Geometry Seminar, Bologna — *2025*

Segre Indices and Lines on Hypersurfaces — **Centre Henri Lebesgue**
Lecture at the Atelier Real and Motivic Geometry, Le Croisic — *2025*

Enumerative Geometry and Physics — **Univeristy of Campinas**
Two Lectures at the University of Campinas Geometry Seminar, Campinas — *2025*

Modularity of open Gromov-Witten invariants for the quintic threefold — **Intern. Centre for Math. Sciences**
Lecture at Theory of Atoms, Educational Workshop, Sofia — *2024*

A "quasi-modularity" for Gromov-Witten invariants for the quintic threefold — **TU Darmstadt**
Lecture at the Seminar on Arithmetic Geometry, Darmstadt — *2024*

Refinements of Donaldson-Thomas Invariants — **Centre Intern. of Math. Meetings**
Lecture at Motivic Homotopy in Interaction, Marseille — *2024*

Lines on hypersurfaces beyond real and complex counts — **LAGARTOS**
Lecture at the LAGARTOS Seminar, Online — *2024*

Arithmetic and motivic refinements of degree zero DT invariants of $\mathbb{A}^3$ — **University of Heidelberg**
Lecture at the conference Hodge theory, Mirror Symmetry, and Physics of Calabi-Yau Moduli , Heidelberg — *2023*

Atyiah-Bott formula and Refined Enumerative Geometry — **Inst. of Pure and Appl. Math.**
Online lecture at the GADEPs Seminar, Rio de Janeiro — *2022*

Gauss–Manin Connection in Disguise and Mirror Symmetry — **Nottingham University**
Online lecture given at the Online Algebraic Geometry Seminar, Nottingham — *2022*

The Fukaya Category and Kontsevich's HMS conjecture — **University of Campinas**
Online lecture given at the Masters Level Geometry Seminar by invitation, Campinas — *2021*

Publications, Preprints and Other Writing

Peer-Reviewed Articles

1. F. Espreafico and J. Walcher, **On Motivic and Arithmetic Refinements of Donaldson-Thomas Invariants**, 2023
   - arXiv:2307.03655
   - **Published** at *Communications in Number Theory and Physics vol. 17 no. 1.*
2. F. Espreafico, **Gauss-Manin Connection in Disguise: Open Gromov-Witten Invariants**, 2022
   - arXiv:2205.08302
   - **Accepted** in *Communications in Mathematical Physics (to appear)*

Preprints

1. F. Espreafico, S. McKean and S. Pauli, **Quadratic Segre Indices**, 2025
   - arXiv:2506.01547

Others

1. F. Espreafico, **Modularity and Enumerative Geometry of Physical Invariants**, 2024
   - PhD Thesis
   - Available here
2. F. Espreafico, **Mirror Symmetry and Fukaya Categories**, 2020
   - Introductory Survey, cited on some nLab articles.
   - Available here (nLab) and here (my page)

Languages

Portuguese: Native
English: General proficiency
German: Conversational
French: Working proficiency

Programming Languages

Python: Good command for basic computational use
Mathematica: Good command
SageMath: Good command

References

Hossein Movasati
Inst. Pure and Appl. Mathematics
hossein@impa.br

Penka Georgieva
Sorbonne University
penka.georgieva@imj-prg.fr

Sabrina Pauli
TU Darmstadt
pauli@mathematik.tu-darmstadt.de

Johannes Walcher
University of Heidelberg
walcher@uni-heidelberg.de

Vinicius G. B. Ramos
Inst. Pure and Appl. Mathematics
vgbramos@impa.br

Nivaldo G. Grulha Junior
University of Sao Paulo
njunior@icmc.usp.br

*PhD advisors in boldface